# Static and kinetic friction



**antocciphysics | TEACHER KEY | CPA + MATCHING SEI**

---

**Learning target:** Distinguish adjustable static friction from sliding friction.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

**01. Warm-up: bring back an earlier idea** *Review: Unit 1, Day 14.*
Ground pushes the runner’s foot forward with 50 N. This external force can accelerate the runner; the partner forces act on different objects.

**02. Worked example: explain the model**
Balanced 3 N pair, then balanced 5 N pair. Friction adjusts while the box remains at rest.

**03. Below the limit**
Actual 8 N left; net 0 N, not 15 N friction.

**04. Crossing the limit**
18>15 N starts slipping; net 18-12=6 N right.

**05. Can friction help?**
Forward. It opposes relative slipping or tendency at the contact, not always the person’s motion.

**06. Review: connect an earlier idea** *Review: Unit 1, Day 5.*
Central position 22 m; interval 20-24 m. Assumes steady velocity and the stated start.

**07. Exit check**
First 12 N opposite push; second 9 N in the direction of the push.

**08. STRETCH: test the model**
4 N: rest possible with 4 N friction. 8 N: limiting equilibrium. 10 N: starts sliding; net 4 N in push direction once moving.

# Static and kinetic friction



**antocciphysics | TEACHER KEY | HONORS + MATCHING SEI**

---

**Learning target:** Distinguish adjustable static friction from sliding friction.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

**01. Warm-up: bring back an earlier idea** *Review: Unit 1, Day 14.*
Equal opposite 15 N forces. Craft acceleration 5 m/s²; asteroid 0.05 m/s² opposite. Compares this interaction's contribution, or assumes other forces negligible.

**02. Worked example: explain the model**
Balanced 3 N pair, then balanced 5 N pair. Friction adjusts while the box remains at rest.

**03. Actual versus maximum**
N 49 N; max 19.6 N; actual 12 N opposite push; a 0.

**04. Sliding case**
$f_k$=0.30×49=14.7 N; net 10.3 N; a 2.06 m/s².

**05. Estimate from data**
Means 8 N and 6 N;$\mu_s$≈0.40 and $\mu_k$≈0.30; static coefficient gives limiting threshold.

**06. Review: connect an earlier idea** *Review: Unit 1, Day 5.*
Central 44 m; half-width 1.3 m; interval [42.7, 45.3] m; full width 2.6 m. Constant velocity and bounded input ranges.

**07. Exit check**
Normal force = 98 N; maximum static friction = 49 N. With a 30 N push: actual friction = 30 N opposite the push and a = 0. With a 60 N push: the box slides; friction = 39.2 N and a = (60 - 39.2)/10 = 2.08 m/s² in the push direction.

**08. STRETCH: test the model**
N=50 N, $f_s$, max=20 N; 22 exceeds it. Kinetic friction 15 N, acceleration 7/5=1.4 m/s². Steady sliding requires 15 N.